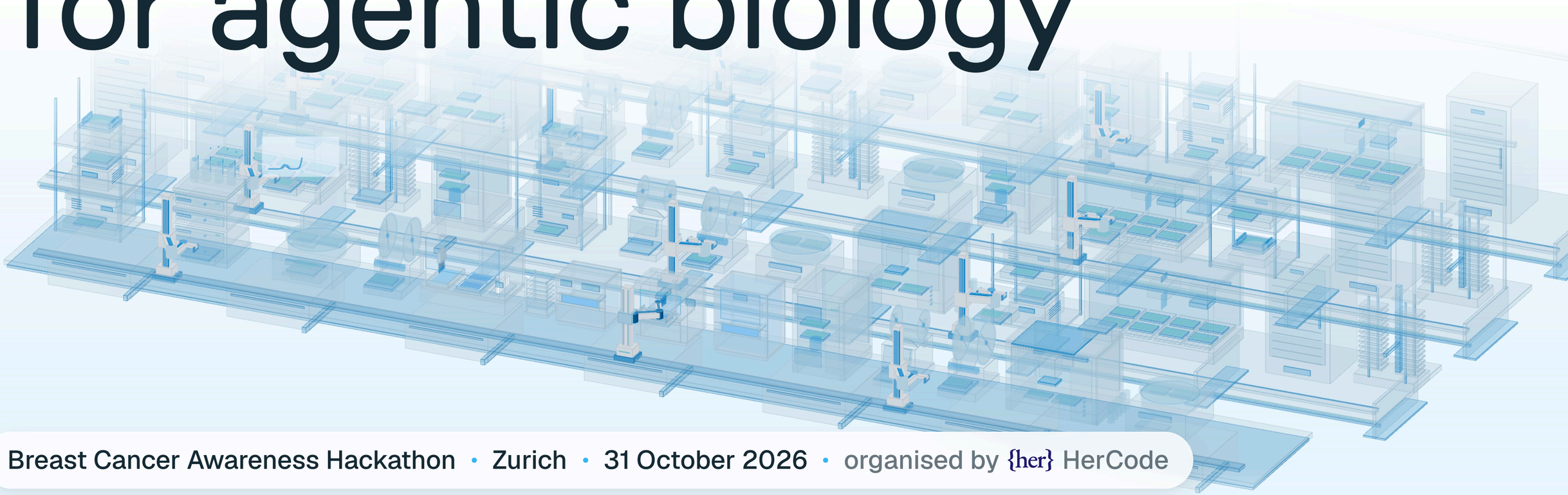
Adaptyv

Gijs de Bruin
GTM TEAM

The automated lab for agentic biology



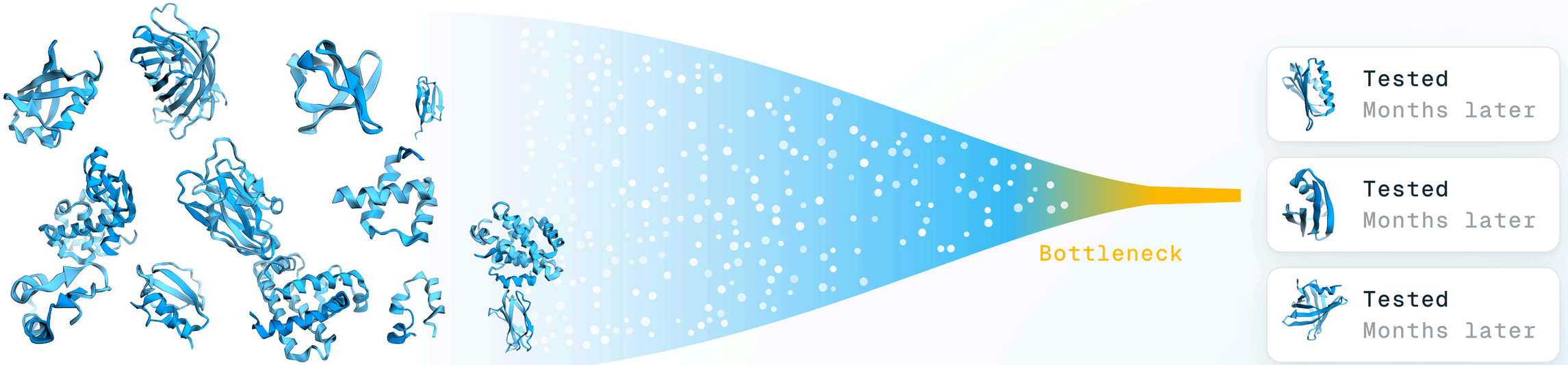
Breast Cancer Awareness Hackathon • Zurich • 31 October 2026 • organised by {her} HerCode

AI can design proteins faster than any lab can test them

Models generate thousands of designs in an afternoon. To know which designs bind, you have to make and measure them. Most designs are never made.

WHY TESTING IS THE BOTTLENECK

- A basic lab costs $2M+ to set up
- Expression and assays need specialist staff
- In-house rounds take months
- Designers want to send sequences by API

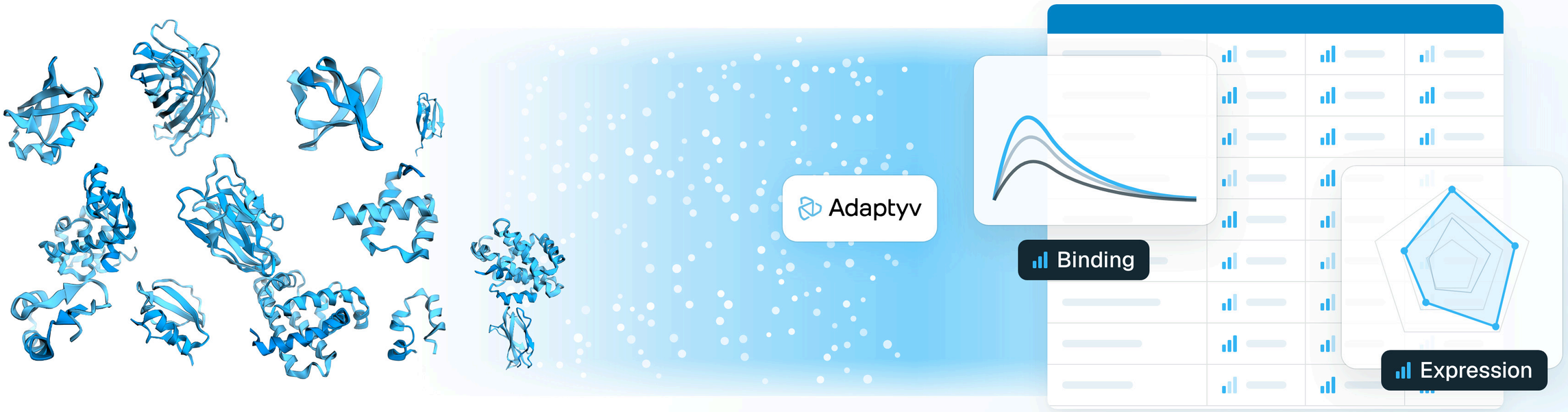


With Adaptyv, every design gets measured

Send the whole batch, not a shortlist. Each design comes back with expression and binding data, ranked by KD.

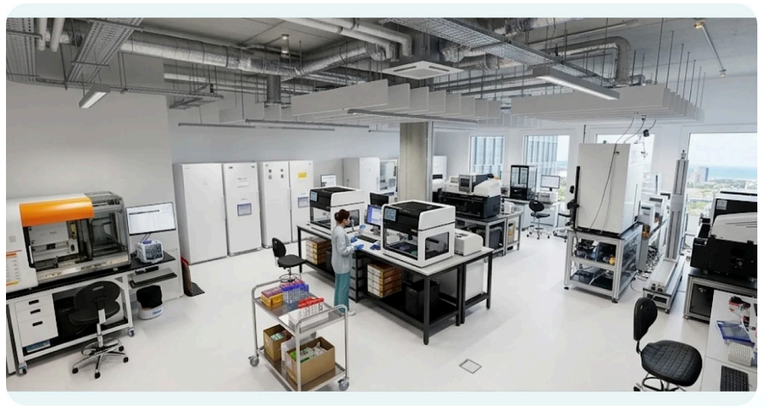
WHAT COMES BACK

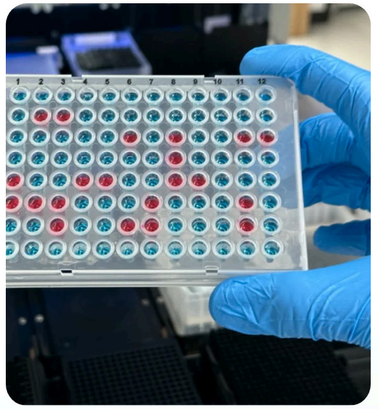
- Every design made and measured
- KD, kon and koff for each binder
- Raw binding curves
- Results by API or in the browser

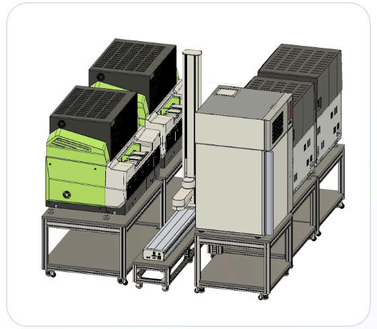


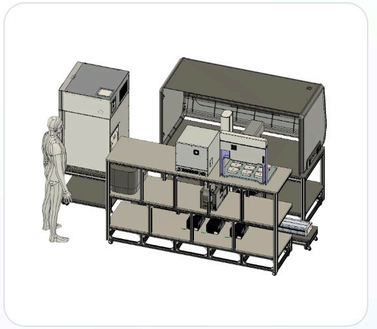
Our first automated lab is live in Switzerland

Robot arms move plates between liquid handlers, cell-free expression and binding instruments. Our software schedules every step, from DNA to binding data.

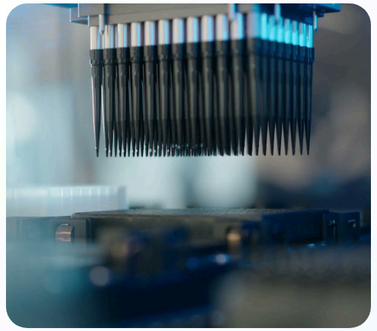










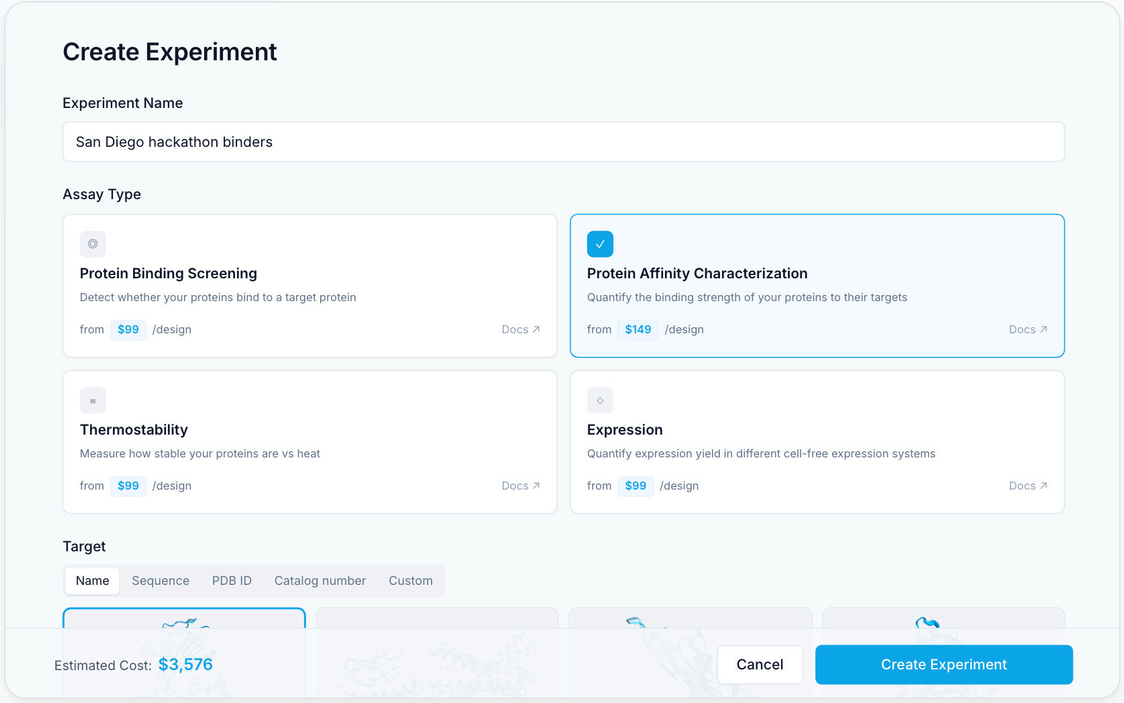




THE ADAPTYV LAB

- Thousands of proteins tested per month
- Workflows automated end to end
- API first, works with AI tools
- Runs 24/7

Submit sequences in the browser or by API



1 · SET UP THE EXPERIMENT

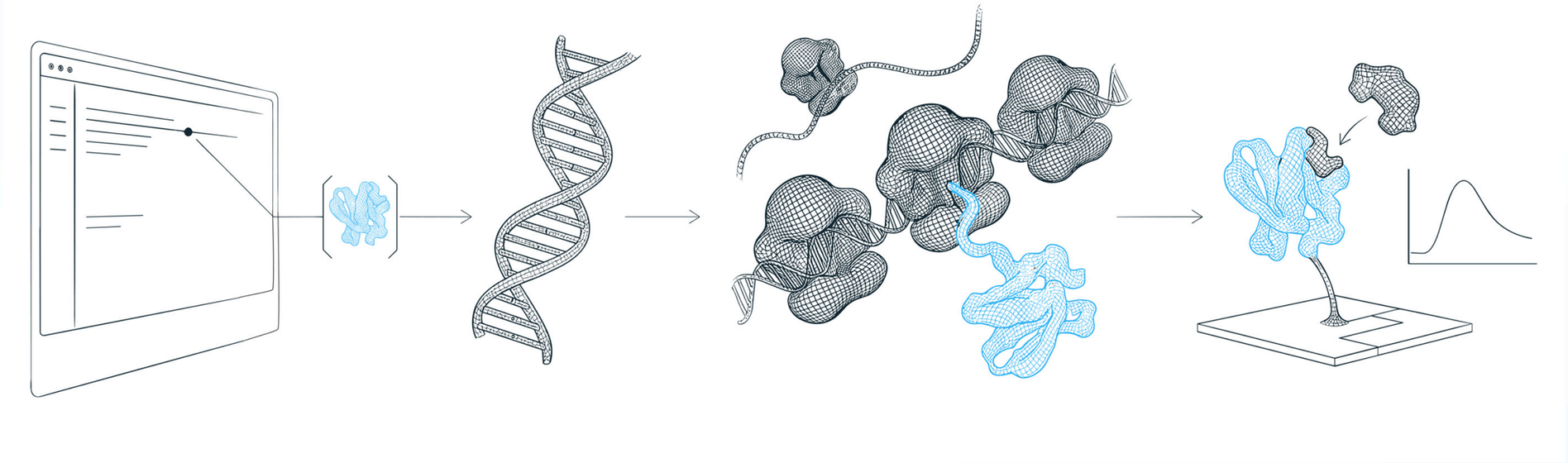
Pick an assay and a target. Upload a CSV, or send the same request to the API. We source the target protein.

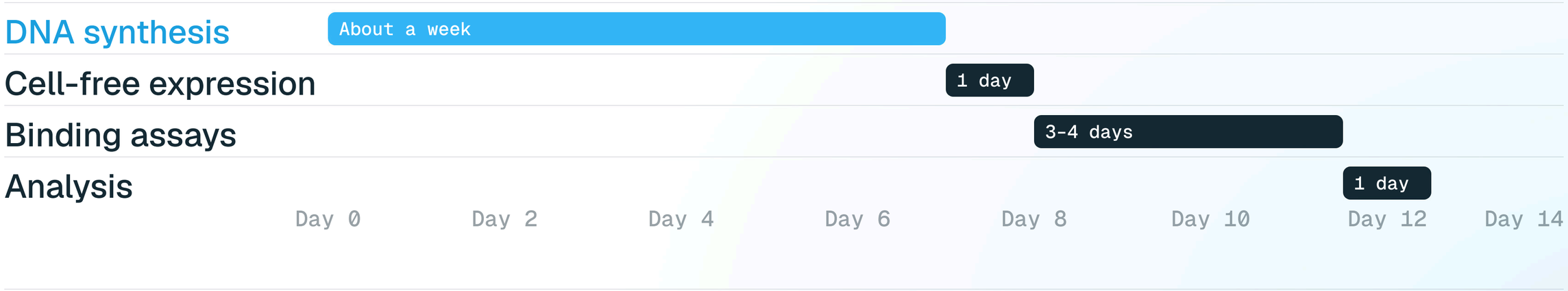


2 · READ THE RESULTS

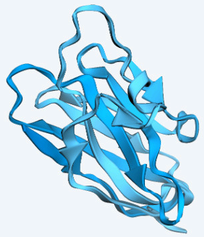
KD, kon and koff for every design, in the browser or as JSON from the API.

We make and measure every design in one automated run

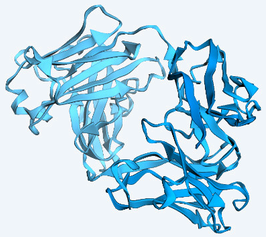




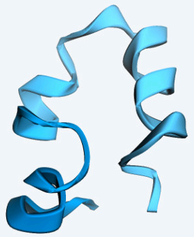
We're on track for measuring 100,000 proteins by end of this year



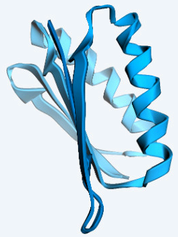
Nanobodies



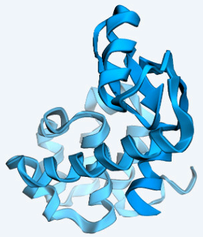
Antibody fragments



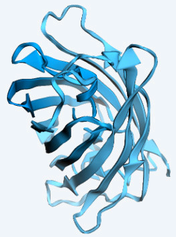
Miniproteins



De novo folds



Enzymes



Fluorescent proteins



The best protein design teams use Adaptyv

Over 170 companies run on Adaptyv, from AI labs to Top 10 pharma.

Chai Discovery

Validated Chai-2, their zero-shot binder model, over several rounds.

10	targets
24	designs per target
2 weeks	per round

ANTHROP\C

Claude ran binder campaigns from one prompt, with no human design decisions.

16	targets
1,320	designs
26.8%	bound

novo nordisk

Benchmarked RFdiffusion and BindCraft, 3× faster than in-house.

4	targets
384	designs per target
3 weeks	per round

Boltz

Validated BoltzGen, their open-source design model, on difficult PDB targets.

12	targets
24	designs per target
6 weeks	end to end

Google DeepMind · Microsoft · EvolutionaryScale · cradle · dyno · DECADE · dreamfold · GenBio

Generate:Biomedicines · Roche · NOVARTIS · MERCK · Massachusetts Institute of Technology · EPFL · HARVARD UNIVERSITY · Stanford

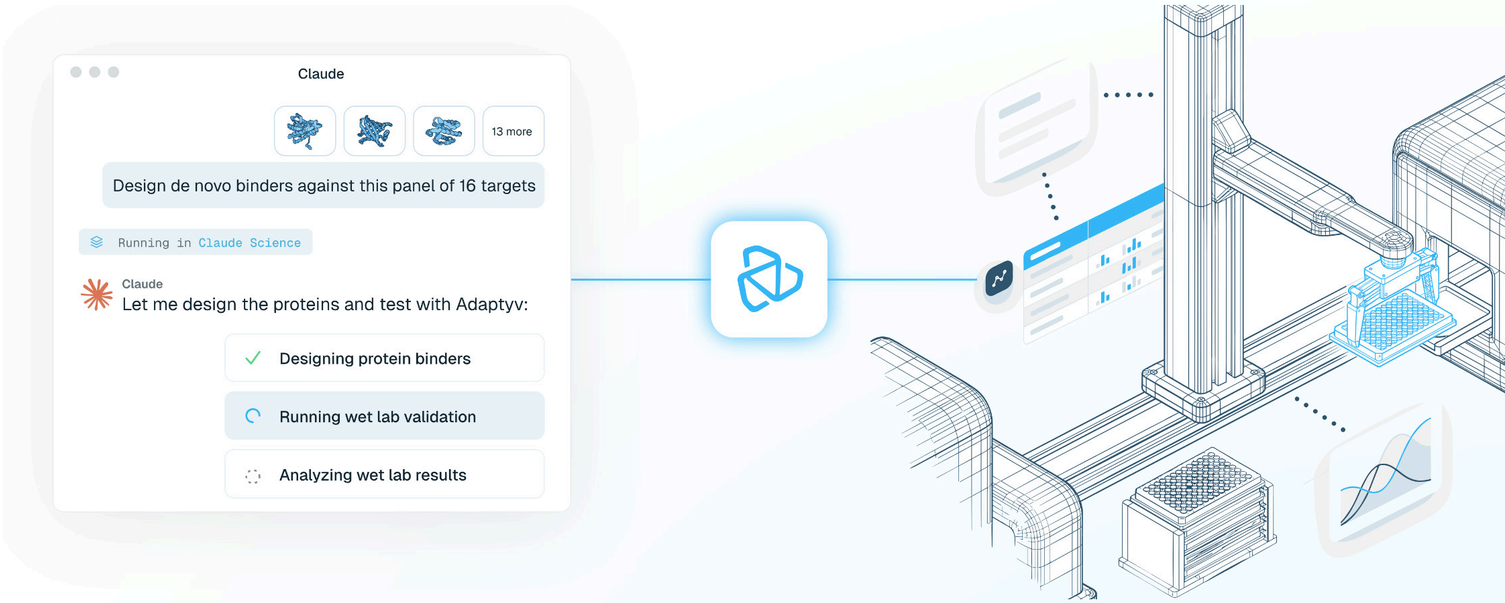
Case study: Testing Claude’s protein designs

ANTHROP\C

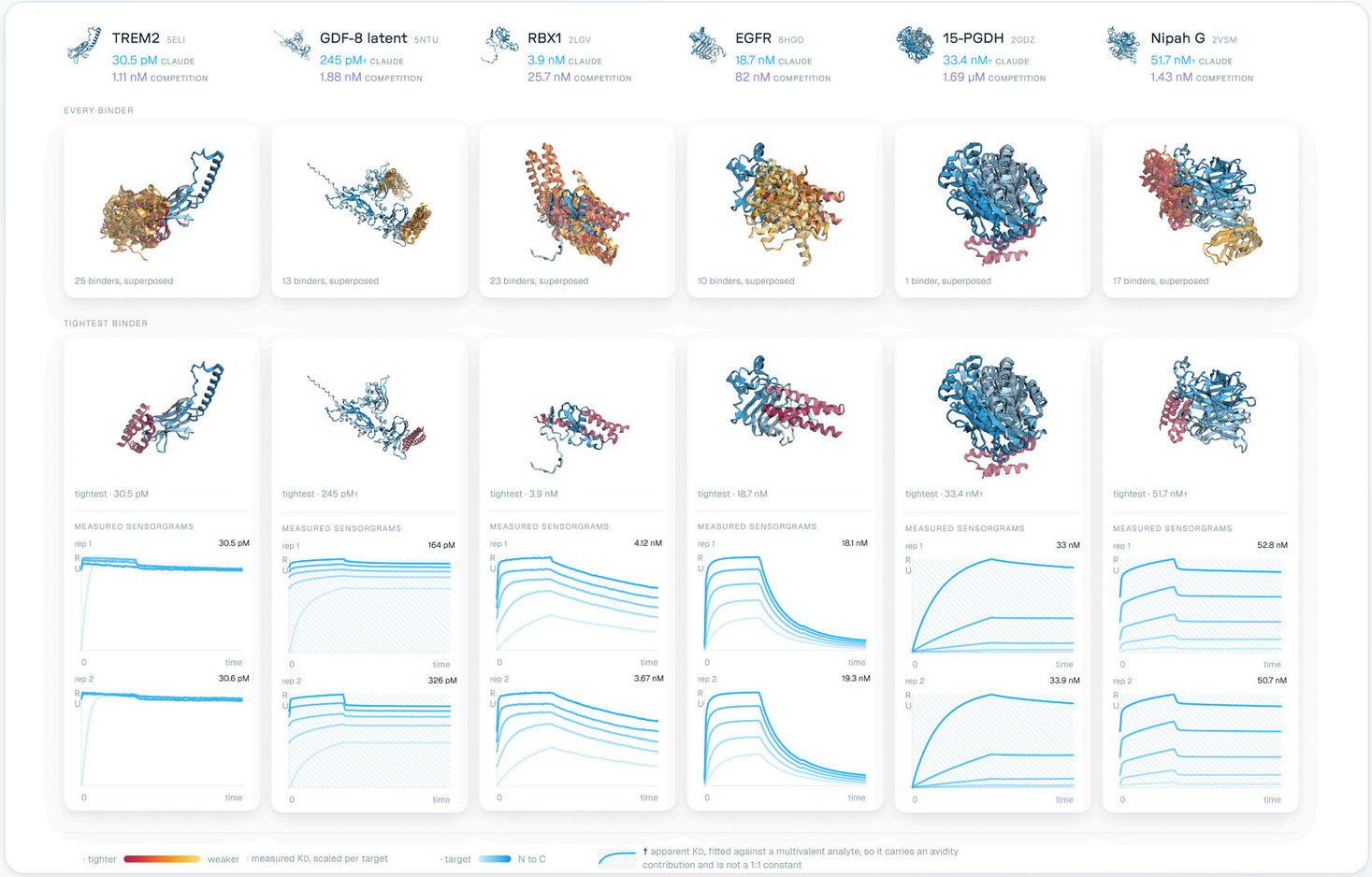
Anthropic gave Claude a target and public design tools in Claude Science. Claude planned the campaign, ran the models and picked the designs. We made each one and measured its binding.

THE CAMPAIGNS

- Claude Mythos Preview and Opus 4.8
- 16 targets, 1,320 designs
- Public design tools only
- SPR, 5 concentrations, in duplicate



Claude matches expert protein designers



THE CAMPAIGNS

Designs tested	1,320
Binders	354
Hit rate	26.8%
Targets with binders	14 of 15
Designs expressed	95%

AGAINST OUR COMPETITION WINNERS

	CLAUDE	COMPETITION
TREM2 hit rate	80%	38%
RBX1 best K_D	3.9 nM	25.7 nM
15-PGDH best K_D	33.4 nM	1.7 µM

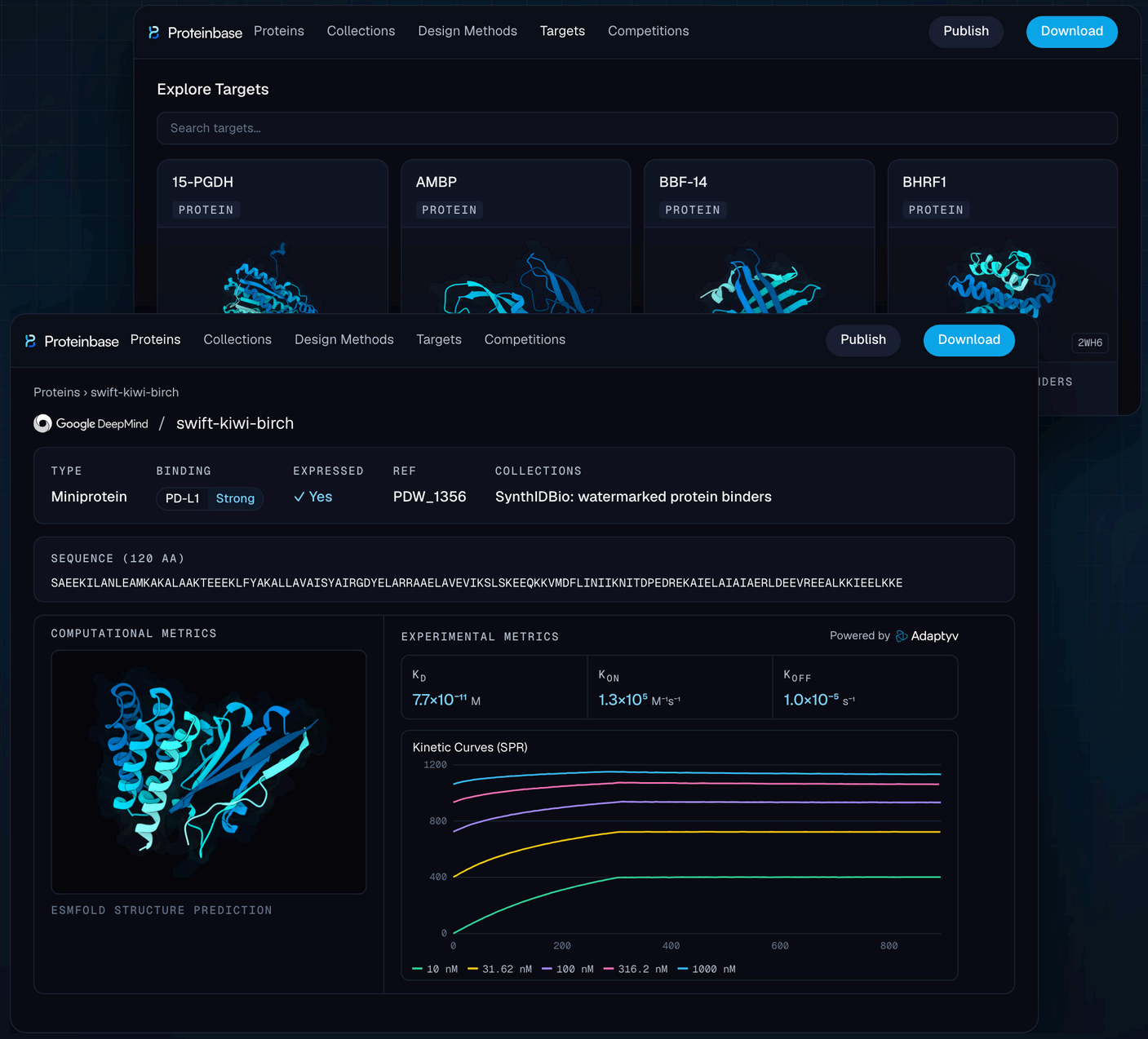
Proteinbase

The open protein data platform

Over 3,000 validated proteins, with every result released on Proteinbase: sequence, target, KD and raw curves.

WHAT YOU CAN DO

- Search by target, design method or lab
- See what already binds your target
- Download sequences and results
- Publish your own data



We're hiring

We are hiring across the board in Lausanne and London!

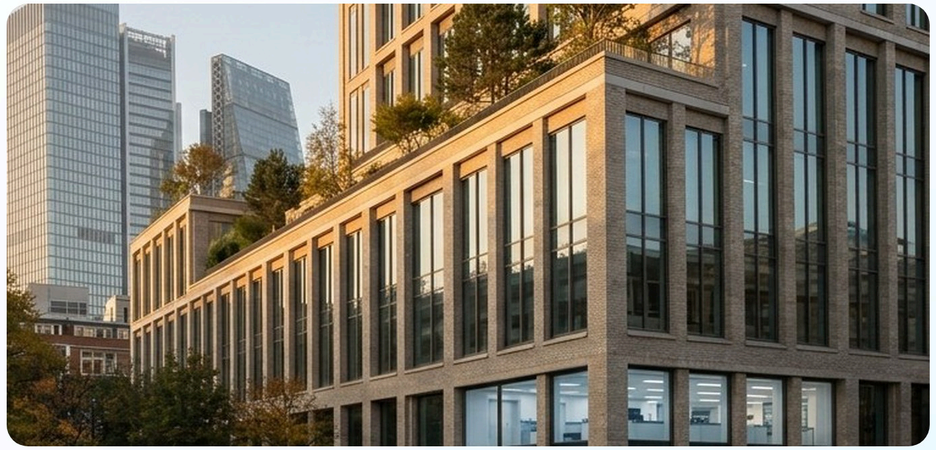
TEAMS WE ARE HIRING INTO

- Biology
- Operations
- Automation
- Software
- Go to market

adaptyvbio.com/careers



Lausanne • production and research



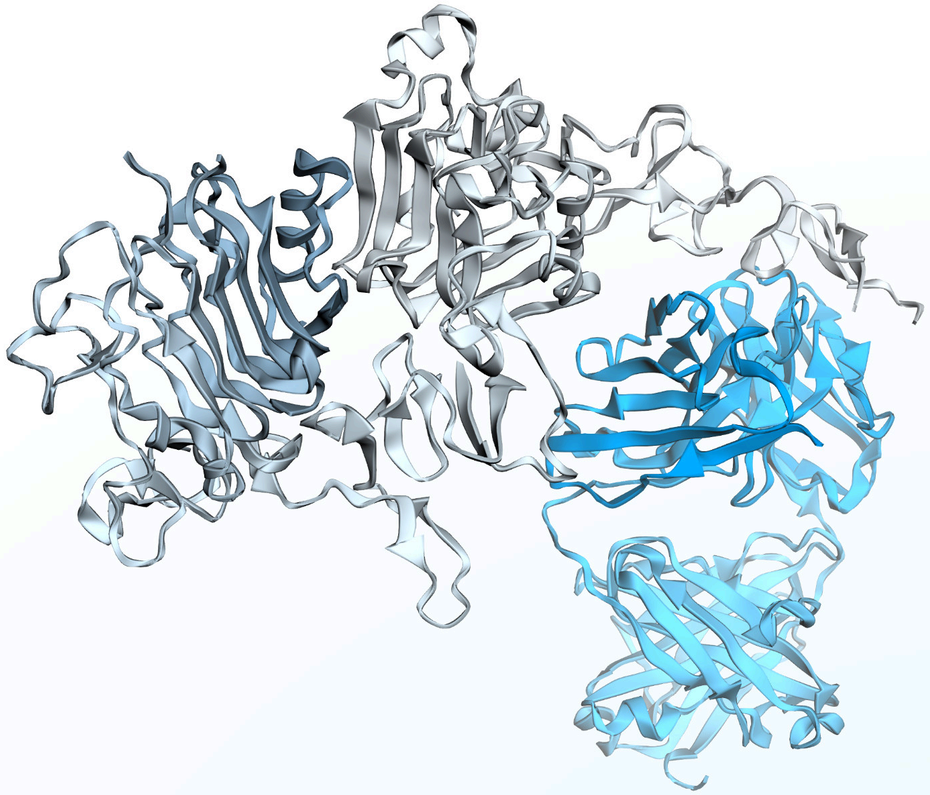
London • research, opening 2027

THE CHALLENGE

Build a better cancer therapeutic

Trastuzumab scFv optimisation on HER2

Breast Cancer Awareness Hackathon • Zurich • 31 October 2026



HER2 with the trastuzumab Fab • PDB 1N8Z

The target: HER2

HER2 is a receptor on the cell surface, a membrane protein that tells the cell to grow and divide. Breast tumours overproduce it in 20–30% of cases, and the cell keeps growing. Current therapeutics target its extracellular domain.



20–30 %

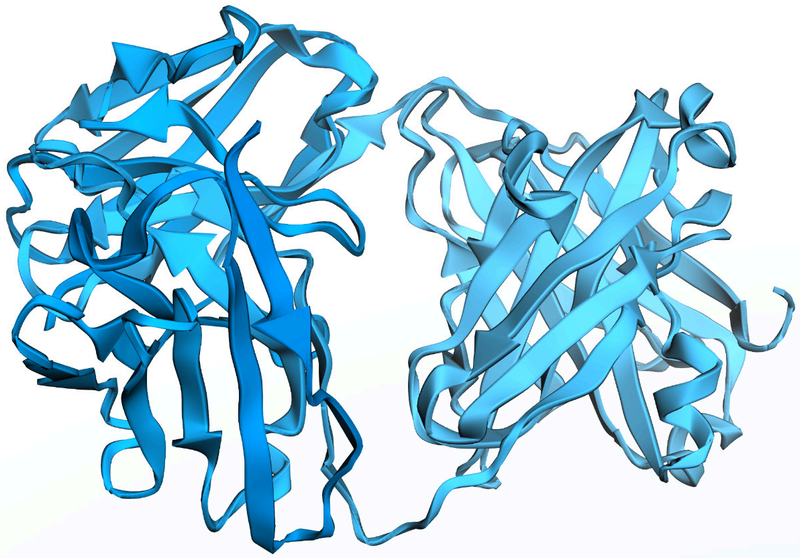
of breast cancers overproduce HER2

630 aa

extracellular domain, Thr 23 – Thr 652

The starting point: trastuzumab scFv

Trastuzumab is one of the main therapies against these tumours. It binds the extracellular domain close to the membrane (PDB 1N8Z, 2.52 Å) and flags the cell for the immune system to kill. Your goal is to improve it as an scFv: 242 aa, heavy chain first, joined to the light chain by a $(GGGGS)_3$ linker.



0.82 nM K_D

the parent, V_H–V_L,
measured by BLI on our own platform

PARENT SEQUENCE, 242 AA

EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGDGFYAMDYWGQGTLVTVSS
GGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGTKVEIK

V_H = PDB 1N8Z chain B, V_L = chain A · the reverse order (V_L, $(GGGGS)_3$, V_H) bound at 1.83 nM

Adaptyv

150 scFvs go into our lab

Pitch your approach to the problem: what models you used, which positions you changed. We assess the ipSAE and the pitch to pick the winner of the challenge today.

FORMAT

Pitch and Q&A

LAB VALIDATION

15 variants per team

45 for the track winner

RESTRICTIONS

At least 5 mutations from the parent. One chain, in the order V_H, $(GGGGS)_3$, V_L, up to 260 aa.

RESULTS

About 3 weeks after the hackathon, public on ProteinBase

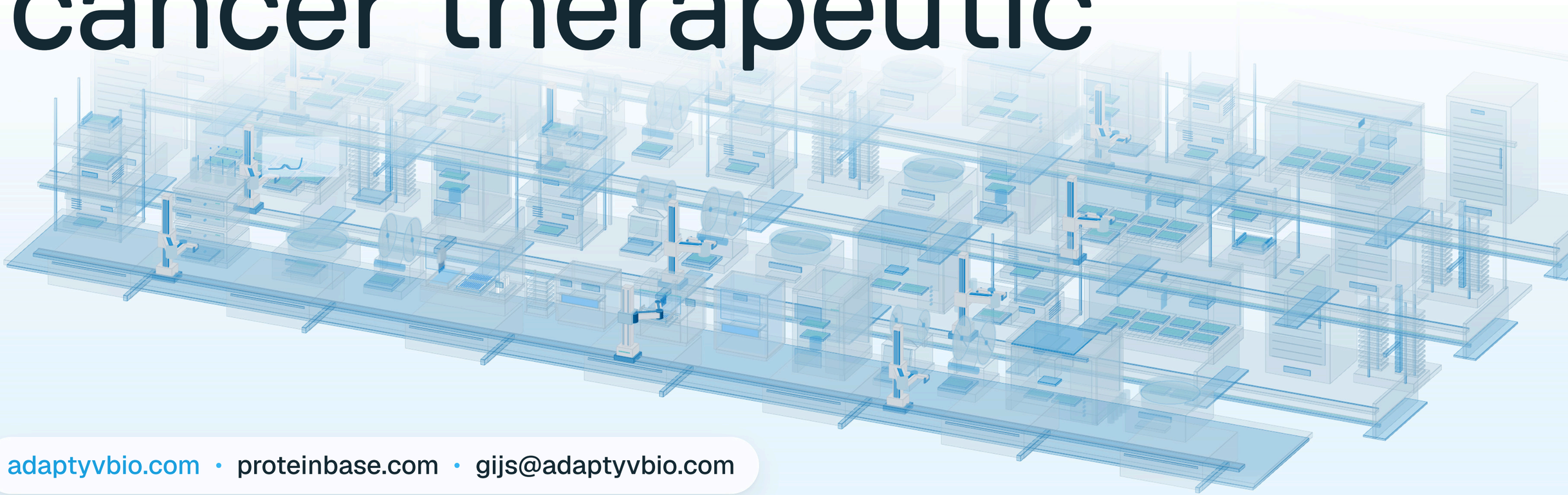
Optimisation toolkit

AbLang2	Paired heavy/light antibody language model. Masked-LM scan across framework and CDRs.
DSM	Synthyra's masked-diffusion ESM2. Sequence-only edits, no structure needed.
ESM2 / ESM3	Zero-shot single- and double-mutant likelihoods over the 242 aa scFv. Free signal before any training.
AbMPNN	Antibody-finetuned ProteinMPNN. Backbone-fixed surface design on the Fab–HER2 complex.
EVOLVEpro	Few-shot active learning over PLM embeddings. A strong baseline when you only get one round.
ProFam-1	Family-as-prompt PLM scoring. Open weights, strong on IgG benchmarks.
Mosaic	Composite objectives over Boltz, AF2 and MPNN. Affinity, expression and developability together.
Boltz-2	Cofold your edited scFv with the HER2 ectodomain, then rank by $ipSAE_{min}$.

proteinbase.com/protein-design-skills — 24 Claude Code skills • adaptyvbio.com/blog/po101 and po102 — optimising on a small test budget

Adaptyv

Let’s build a better cancer therapeutic



adaptyvbio.com • proteinbase.com • gijs@adaptyvbio.com

Improve the trastuzumab scFv against HER2

THE TARGET

HER2 extracellular domain, 630 aa, Thr 23 – Thr 652, UniProt P04626.

PDB 1N8Z is the whole complex: the HER2 ectodomain with the trastuzumab Fab, at 2.52 Å. Trastuzumab binds the part closest to the membrane.

THE PARENT

242 aa single chain: V_H, $(GGGGS)_3$ linker, V_L. The two domains are **PDB 1N8Z** chains B and A. Sequence on the starting-point slide.

0.82 nM K_D, measured by BLI on our own platform. That is the number to beat.

RULES

At least **5 mutations** from the parent. One chain, in the order V_H, $(GGGGS)_3$, V_L, up to 260 aa.

JUDGED ON

A creative approach to design optimisation. The tools you used and why. The ipSAE of your pool.

RESULTS

15 variants per team go to the lab, 45 for the winner. All 150 are made, expressed and measured in Lausanne. You get K_D, k_{on}, k_{off} and the raw sensorgrams about 3 weeks after, public on ProteinBase.

USEFUL LINKS

proteinbase.com/protein-design-skills
24 protein design skills for Claude Code.

adaptyvbio.com/blog/po101 and po102
Optimising on a small test budget.

huggingface.co/datasets/Anthropic/claude-protein-binder-design
Anthropic's own campaign prompt, at **prompts/single_target/EGFR.md**. Written for EGFR, but a good template.

OPTIMISATION TOOLKIT

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Adaptyv